

MAC21 — The Complete 2-Mark Bank

Every 2-mark question that has been asked, plus every one the syllabus can support. 76 questions with model answers. You face five 2-markers and choose the easier of two in each unit — know most of this and all **10 marks are free**.

4x

= times it has actually appeared across the 6 papers ·

LIKELY

 = not yet asked, but squarely on the syllabus and answerable in two lines.

Part (a) is a definition, a formula to state, or "write the steps" — never a calculation (the two exceptions are marked).
Answers below are exam-length: write roughly this much, no more.

UNIT I — Series Expansion, Newton-Raphson, Maxima-Minima

#	QUESTION	ANSWER	SEEN
1	Write Taylor's series for a function of one variable	$f(x) = f(a) + (x-a)f'(a)/1! + (x-a)^2f''(a)/2! + \dots + (x-a)^nf^{(n)}(a)/n! + \dots$ (about the point $x = a$)	1x
2	Write Maclaurin's series for a function of one variable	$f(x) = f(0) + xf'(0)/1! + x^2f''(0)/2! + x^3f'''(0)/3! + \dots$ (Taylor's series about $x = 0$)	2x
3	State Taylor's series for two variables	$f(x,y) = f(a,b) + [hf_x + kf_y] + (1/2!)[h^2f_{xx} + 2hkf_{xy} + k^2f_{yy}] + \dots$ where $h = x-a$, $k = y-b$, all derivatives evaluated at (a,b) .	2x
4	State Maclaurin's series for two variables	Same as above with $a = b = 0$: $f(x,y) = f(0,0) + [xf_x + yf_y] + (1/2!)[x^2f_{xx} + 2xyf_{xy} + y^2f_{yy}] + \dots$, derivatives at $(0,0)$.	LIKELY
5	Define algebraic and transcendental equations , with an example each	Algebraic: contains only algebraic terms — powers of x . e.g. $x^3 - x^2 + 9 = 0$. Transcendental: contains exponential, logarithmic or trigonometric terms. e.g. $x e^x - 9 = 0$, $x \sin x + \log x = 0$.	2x
6	Write the Newton-Raphson iteration formula	$x_{n+1} = x_n - f(x_n)/f'(x_n)$, $n = 0, 1, 2, \dots$	LIKELY
7	Geometrical interpretation of Newton-Raphson	Draw the tangent to $y = f(x)$ at the point $(x_n, f(x_n))$. The point where that tangent cuts the x -axis is the next approximation x_{n+1} . Repeating drives the point towards the root.	2x
8	Explain N-R for a system of non-linear simultaneous equations	For $f(x,y)=0$, $g(x,y)=0$: at (x_0, y_0) solve the linear pair $-f = hf_x + kf_y$, $-g = hg_x + kg_y$ for the corrections h, k ; then $x_1 = x_0+h$, $y_1 = y_0+k$. Repeat.	1x
9	Can N-R find a root of $x e^{-x} = 0$ near $x = 2$? Justify	No. $f(x) = x e^{-x}$ has $f'(x) = e^{-x}(1-x)$, which is zero at $x = 1$ and very small near $x = 2$. The iteration divides by f' , so it diverges. N-R fails when f' is zero or near zero at the starting point.	1x
10	State the intermediate value property (how to locate a root)	If f is continuous on $[a,b]$ and $f(a), f(b)$ have opposite signs , then $f(x)=0$ has at least one real root c in (a,b) .	LIKELY
11	Define maxima and minima for a function of two variables	$f(a,b)$ is a maximum if $f(a+h, b+k) < f(a,b)$ for all small h,k ; a minimum if $f(a+h, b+k) > f(a,b)$. Both are <i>local</i> (relative) extrema.	1x
12	State the working rule (r-s-t test) for extrema of $f(x,y)$	Solve $f_x=0$, $f_y=0$ for stationary points. Put $r=f_{xx}$, $s=f_{xy}$, $t=f_{yy}$. $rt-s^2>0$, $r<0 \rightarrow$ maximum · $rt-s^2>0$, $r>0 \rightarrow$ minimum · $rt-s^2<0 \rightarrow$ saddle point · $rt-s^2=0 \rightarrow$ test fails.	LIKELY
13	Define a saddle point	A stationary point that is neither a maximum nor a minimum — the function increases along one direction and decreases along another. Occurs when $rt - s^2 < 0$.	LIKELY
14	State Lagrange's method of undetermined multipliers	To find extrema of $u(x,y,z)$ subject to $\phi(x,y,z) = 0$, form $F = u + \lambda\phi$ and solve $F_x = 0$, $F_y = 0$, $F_z = 0$ together with $\phi = 0$. λ is the undetermined multiplier.	LIKELY
15	Write two differences between analytical and numerical methods	Analytical: gives an exact closed-form solution; applies only to a limited class of equations. Numerical: gives an approximate solution to a stated accuracy; applies to almost any equation, and is iterative/computational.	1x

UNIT II — First Order ODEs, Applications, Numerical Solution

#	QUESTION	ANSWER	SEEN
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16	Define orthogonal trajectories	Two families of curves such that every member of one family cuts every member of the other at right angles. Each family is the orthogonal trajectory of the other.	2x
17	Steps to find orthogonal trajectories of $f(x, y, c) = 0$ (Cartesian)	1. Differentiate w.r.t. x and eliminate the parameter c to get the family's DE. 2. Replace dy/dx by $-dx/dy$. 3. Solve the resulting DE — that is the orthogonal family.	1x
18	Steps for orthogonal trajectories in polar form	1. Differentiate $f(r, \theta, c) = 0$ w.r.t. θ and eliminate c . 2. Replace $dr/d\theta$ by $-r^2(d\theta/dr)$. 3. Solve the resulting DE.	1x
19	Write the integrating factor for a linear first-order DE	For $dy/dx + P(x) \cdot y = Q(x)$: I.F. = $e^{\int P dx}$, and the solution is $y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dx + c$.	LIKELY
20	Write the DE of an L-R circuit	$L (di/dt) + R i = E \rightarrow di/dt + (R/L) i = E/L$	LIKELY
21	Write the DE of an R-C circuit with source E	$R (dq/dt) + q/C = E \rightarrow dq/dt + q/(CR) = E/R$, with $i = dq/dt$.	1x
22	Write the DE of a closed L-C circuit in series, no applied e.m.f.	$L (d^2q/dt^2) + q/C = 0$ (no resistance, no source — simple harmonic)	2x
23	Write the DE of an L-C-R circuit	$L (di/dt) + R i + q/C = E \rightarrow di/dt + (R/L) i + q/(LC) = E/L$	LIKELY
24	State Newton's law of cooling	The rate of change of temperature of a body is proportional to the difference between its temperature and that of the surrounding medium: $dT/dt = -k(T - T_0)$, giving $T = T_0 + (T_1 - T_0)e^{-kt}$.	LIKELY
25	Write Euler's formula to solve a first order, first degree ODE	For $dy/dx = f(x, y)$, $y(x_0) = y_0$: $y_{n+1} = y_n + h \cdot f(x_n, y_n)$, h = step length.	2x
26	Write the modified Euler formula	Predictor: $y_{n+1}^{(0)} = y_n + h f(x_n, y_n)$ Corrector: $y_{n+1}^{(k+1)} = y_n + (h/2)[f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(k)})]$, iterated until it settles.	LIKELY
27	Write the Runge-Kutta formula of fourth order	$k_1 = h f(x_0, y_0)$ · $k_2 = h f(x_0 + h/2, y_0 + k_1/2)$ · $k_3 = h f(x_0 + h/2, y_0 + k_2/2)$ · $k_4 = h f(x_0 + h, y_0 + k_3)$ $y_1 = y_0 + (1/6)(k_1 + 2k_2 + 2k_3 + k_4)$	1x
28	Write the Taylor's series method formula for an ODE	$y(x) = y_0 + (x-x_0)y'_0/1! + (x-x_0)^2 y''_0/2! + (x-x_0)^3 y'''_0/3! + \dots$ Derivatives are found by repeatedly differentiating $y' = f(x, y)$ and evaluating at (x_0, y_0) .	LIKELY

UNIT III — Linear Differential Equations of Higher Order

#	QUESTION	ANSWER	SEEN
29	Define linear and non-linear differential equations , with examples	Linear: the dependent variable and all its derivatives occur only in the first degree, and no products of them appear. e.g. $y'' + 3y' + 2y = e^x$. Non-linear: any other — a power, product, or non-linear function of y or its derivatives. e.g. $y'' + y^2 = 0$ or $y y' = x$.	3x
30	Define a non-homogeneous LDE of second order , with an example	A second-order linear DE whose right-hand side is not zero: $a_0 y'' + a_1 y' + a_2 y = \phi(x)$ with $\phi(x) \neq 0$. e.g. $y'' + 4y' + 4y = e^{2x}$. (If $\phi(x) = 0$ it is homogeneous.)	1x
31	Define complementary function and particular integral	C.F. = the general solution of the homogeneous equation $f(D)y = 0$, carrying the arbitrary constants. P.I. = any one solution of $f(D)y = \phi(x)$, given by $(1/f(D))\phi(x)$. Complete solution $y = \text{C.F.} + \text{P.I.}$	LIKELY
32	Write the C.F. if the roots of the A.E. are $-3 \pm i, 2, 2, 2$	$y^c = e^{-3x}(C_1 \cos x + C_2 \sin x) + (C_3 + C_4 x + C_5 x^2)e^{2x}$	1x
33	Write the C.F. of a 4th-order LDE if two roots are imaginary and two are real and distinct	Roots $\alpha \pm i\beta, m_1, m_2$: $y^c = e^{m_1 x}(C_1 \cos \beta x + C_2 \sin \beta x) + C_3 e^{m_2 x} + C_4 e^{m_3 x}$	1x
34	Rules for the C.F. from the roots (general)	Real distinct $m_1, m_2 \rightarrow C_1 e^{m_1 x} + C_2 e^{m_2 x}$ · repeated m (k times) $\rightarrow (C_1 + C_2 x + \dots + C_k x^{k-1})e^{m x}$ · complex $\alpha \pm i\beta \rightarrow e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$.	LIKELY
35	Write the steps for solving Cauchy's LDE	1. Put $x = e^z$, i.e. $z = \log x$; write $D = d/dz$. 2. Use $x y' = Dy$, $x^2 y'' = D(D-1)y$, $x^3 y''' = D(D-1)(D-2)y$. 3. The equation becomes constant-coefficient in z — find C.F. and P.I. 4. Substitute back $z = \log x$.	2x

36	Write the steps for solving Legendre's LDE	Same as Cauchy's, but put $ax + b = e^z$ and use $(ax+b)y' = aDy$, $(ax+b)^2 y'' = a^2 D(D-1)y$, $(ax+b)^3 y''' = a^3 D(D-1)(D-2)y$. <i>Cauchy's is the special case $a=1$, $b=0$.</i>	LIKELY
37	Write the steps for the method of variation of parameters	1. Put the equation in standard form $y'' + Py' + Qy = R$ and find the C.F. $C_1 y_1 + C_2 y_2$. 2. Compute the Wronskian $W = y_1 y_2' - y_2 y_1'$. 3. $y_v = -y_1 \int (y_2 R/W) dx + y_2 \int (y_1 R/W) dx$. 4. $y = C.F. + y_v$.	1×
38	Define an initial value problem	A DE together with conditions on the unknown function and its derivatives all specified at one and the same value of the independent variable. e.g. $y'' + 2y' + y = 0$, $y(0)=1$, $y'(0)=2$.	LIKELY
39	Define a boundary value problem, with an example	A DE with conditions specified at two or more different values of the independent variable. e.g. $y'' + y = 0$, $y(0)=2$, $y(\pi/2)=-2$.	2×
40	Write the P.I. rule for e^{ax}	$(1/f(D))e^{ax} = e^{ax}/f(a)$, provided $f(a) \neq 0$. If $f(a)=0$, multiply by x and differentiate the denominator: $x e^{ax}/f'(a)$; repeat if needed.	LIKELY
41	Write the P.I. rule for $\sin ax$ or $\cos ax$	In $(1/f(D^2))$ replace D^2 by $-a^2$. If the denominator becomes zero, multiply by x and differentiate the denominator.	LIKELY
42	Write the exponential shift rule (P.I. for $e^{ax} V(x)$)	$(1/f(D)) \cdot e^{ax} V(x) = e^{ax} \cdot (1/f(D+a)) \cdot V(x)$ <i>Every D becomes $D+a$ — this is the step most often got wrong.</i>	LIKELY
43	If $D = d/dx$, prove $(1/(D+a))X = e^{-ax} \int X e^{ax} dx$	Let $y = (1/(D+a))X$, so $(D+a)y = X$, i.e. $y' + ay = X$. This is linear with I.F. e^{ax} : $(y e^{ax})' = X e^{ax}$. Integrating, $y e^{ax} = \int X e^{ax} dx$, hence $y = e^{-ax} \int X e^{ax} dx$. ■ (Usually a 4-marker — appeared 4x.)	4×

UNIT IV — Interpolation, Numerical Differentiation & Integration

#	QUESTION	ANSWER	SEEN
44	Define forward and backward difference, with an example	Forward: $\Delta y_n = y_{n+1} - y_n$. Backward: $\nabla y_n = y_n - y_{n-1}$. e.g. for $y = 2, 5, 10$: $\Delta y_0 = 3$, $\Delta y_1 = 5$; $\nabla y_2 = 5$.	1×
45	Obtain $\Delta^2 y_n$ in terms of y values	$\Delta^2 y_n = \Delta(\Delta y_n) = \Delta(y_{n+1} - y_n) = (y_{n+2} - y_{n+1}) - (y_{n+1} - y_n)$ $\therefore \Delta^2 y_n = y_{n+2} - 2y_{n+1} + y_n$	2×
46	Obtain $\nabla^2 y_n$ in terms of y values	$\nabla^2 y_n = y_n - 2y_{n-1} + y_{n-2}$	1×
47	Define the shift operator E and its relation to Δ and ∇	$E y_n = y_{n+1}$. Then $\Delta = E - 1$ and $\nabla = 1 - E^{-1}$, so $E = 1 + \Delta$.	LIKELY
48	Determine $\Delta^2 y_2$ and $\Delta^3 y_1$ from $y = 0.0567, 1.2345, 1.5678, 1.0023, 1.786$ (a rare calculation)	Δ : 1.1778, 0.3333, -0.5655, 0.7837 Δ^2 : -0.8445, -0.8988, 1.3492 Δ^3 : -0.0543, 2.2480 $\Delta^2 y_2 = -0.8988$, $\Delta^3 y_1 = -0.0543$	1×
49	Write Newton-Gregory forward interpolation formula	$y = y_0 + p\Delta y_0 + p(p-1)/2! \cdot \Delta^2 y_0 + p(p-1)(p-2)/3! \cdot \Delta^3 y_0 + \dots$, where $p = (x - x_0)/h$.	LIKELY
50	Write Newton-Gregory backward interpolation formula	$y = y_n + p\nabla y_n + p(p+1)/2! \cdot \nabla^2 y_n + p(p+1)(p+2)/3! \cdot \nabla^3 y_n + \dots$, where $p = (x - x_n)/h$.	LIKELY
51	Write the formula for $y'(x)$ and $y''(x)$ using Newton-Gregory forward	$dy/dx = (1/h)[\Delta y_0 + (2p-1)/2 \cdot \Delta^2 y_0 + (3p^2-6p+2)/6 \cdot \Delta^3 y_0 + \dots]$ $d^2y/dx^2 = (1/h^2)[\Delta^2 y_0 + (p-1)\Delta^3 y_0 + (12p^2-36p+22)/24 \cdot \Delta^4 y_0 + \dots]$ At $x = x_0$ ($p=0$): $y' = (1/h)[\Delta y_0 - \Delta^2 y_0/2 + \Delta^3 y_0/3 - \dots]$	2×
52	Write Lagrange's interpolation formula	$y = \sum_i y_i \cdot \prod_{j \neq i} (x - x_j)/(x_i - x_j)$ — valid for unequally spaced data.	LIKELY
53	Write Lagrange's inverse interpolation formula for two points	$x = x_0(y - y_1)/(y_0 - y_1) + x_1(y - y_0)/(y_1 - y_0)$ <i>Same formula with the roles of x and y interchanged.</i>	2×
54	Write Newton's divided difference formula	$y = y_0 + (x-x_0)f[x_0, x_1] + (x-x_0)(x-x_1)f[x_0, x_1, x_2] + \dots$ where $f[x_0, x_1] = (y_1 - y_0)/(x_1 - x_0)$ and $f[x_0, x_1, x_2] = (f[x_1, x_2] - f[x_0, x_1])/(x_2 - x_0)$.	LIKELY
55	Find the first order divided difference of $x_0=1$, $x_1=2$ if $y = x^3 + 4x + 3$ (a rare calculation)	$y(1) = 1+4+3 = 8$, $y(2) = 8+8+3 = 19$ $f[x_0, x_1] = (19 - 8)/(2 - 1) = 11$	1×
56	Write the trapezoidal rule	$\int y dx = (h/2)[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$ — any number of intervals.	LIKELY
57	Write Simpson's 1/3rd rule	$\int y dx = (h/3)[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots) + 2(y_2 + y_4 + y_6 + \dots)]$ Requires n to be even .	2×

58	Write Simpson's 3/8th rule	$\int y \, dx = (3h/8) [(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \dots) + 2(y_3 + y_6 + \dots)]$ Requires n to be a multiple of 3 . The 2 multiplies every third ordinate.	LIKELY
59	State the Newton-Cotes quadrature formula	The general formula obtained by integrating Newton's forward interpolation polynomial over $[x_0, x_0 + nh]$. Trapezoidal ($n=1$), Simpson's 1/3 ($n=2$) and Simpson's 3/8 ($n=3$) are its special cases.	LIKELY
60	Relation between ordinates and intervals	Number of intervals n = number of ordinates $- 1$. e.g. "seven equidistant ordinates" means $n = 6$, and $h = (b-a)/n$.	LIKELY

UNIT V — Linear Algebra

#	QUESTION	ANSWER	SEEN
61	Define row echelon form of a matrix	A matrix is in row echelon form if: (i) all zero rows are below every non-zero row; (ii) the first non-zero entry of each row (the leading entry) lies strictly to the right of the leading entry of the row above.	1×
62	Define the rank of a matrix	The number of non-zero rows when the matrix is reduced to echelon form — equivalently the order of its largest non-vanishing minor. Denoted $\rho(A)$, and $\rho(A) = \rho(A^T)$.	1×
63	Write the conditions for consistency and inconsistency of $AX = B$	Consistent if $\rho(A) = \rho(A B)$: $= n$ (number of unknowns) $\rightarrow$ unique solution $= r < n \rightarrow$ infinitely many, with $n-r$ free parameters Inconsistent (no solution) if $\rho(A) \neq \rho(A B)$.	1×
64	Geometrical interpretation of infinitely many solutions for $2x+y=3$, $4x+2y=6$	The second equation is exactly twice the first, so both represent the same straight line . The lines coincide, so every point on the line is a solution — hence infinitely many.	2×
65	Geometrical interpretation of a unique solution / no solution	Unique: the lines intersect at exactly one point. No solution: the lines are parallel and distinct — they never meet.	LIKELY
66	Define elementary row transformations	(i) Interchange two rows; (ii) multiply a row by a non-zero scalar; (iii) add a multiple of one row to another. They do not change the rank.	LIKELY
67	Trivial and non-trivial solutions of $AX = 0$	Trivial: $X = 0$, which always exists. Non-trivial solutions exist only if $ A = 0$, i.e. $\rho(A) < n$.	LIKELY
68	Define eigenvalue and eigenvector	For a square matrix A , if there is a scalar λ and a non-zero vector X with $AX = \lambda X$, then λ is an eigenvalue of A and X the corresponding eigenvector.	LIKELY
69	Write the characteristic equation	$ A - \lambda I = 0$. Its roots are the eigenvalues. <i>Note: the eigenvector then comes from $(A - \lambda I)X = 0$ — a different equation.</i>	LIKELY
70	Two properties of eigenvalues	Sum of the eigenvalues = trace of A ; product of the eigenvalues = $ A $. Also: A and A^T have the same eigenvalues, and if λ is an eigenvalue of A then $1/\lambda$ is one of A^{-1} .	LIKELY
71	Define modal and spectral matrix	Modal matrix P: the matrix whose columns are the eigenvectors of A — it diagonalizes A . Spectral matrix D: the resulting diagonal matrix, whose diagonal entries are the eigenvalues in the same order.	1×
72	Define similarity transformation	If P is non-singular, $P^{-1}AP = D$ is a similarity transformation. Similar matrices have the same eigenvalues.	LIKELY
73	Is every square matrix diagonalizable? Justify	No. An $n \times n$ matrix is diagonalizable only if it has n linearly independent eigenvectors . A repeated eigenvalue may not supply enough of them — e.g. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has $\lambda=1$ twice but only one independent eigenvector, so it is not diagonalizable.	1×
74	What does Rayleigh's power method determine?	The numerically largest (dominant) eigenvalue of a square matrix, and its corresponding eigenvector, by repeatedly forming $Y = AX$ and normalising by the largest component.	LIKELY
75	State the condition for convergence of Gauss-Seidel (diagonal dominance)	In each row the magnitude of the diagonal element must be at least the sum of the magnitudes of the other elements in that row: $ a_{ii} \geq \sum_{j \neq i} a_{ij} $. Reorder the equations to achieve this before iterating.	LIKELY
76	Difference between Gauss elimination and Gauss-Seidel	Gauss elimination is a <i>direct</i> method — row-reduce to triangular form and back-substitute, giving the exact solution in finitely many steps. Gauss-Seidel is an <i>iterative</i> method — start from a guess and refine repeatedly; needs diagonal dominance to converge.	LIKELY

How to use this

1. Cover the answer column and go down the list. Anything you cannot say in one line, mark it.
2. Second pass on the marked ones only.
3. Priority if time is short: the **2×** and **3×** rows first, then **1×**, then LIKELY.
4. In the hall, **do all five 2-markers before anything else** — 10 marks in about 15 minutes, while you are still calm, and you choose the easier of two in each unit.

Write two or three lines, not an essay. A definition plus an example, or the formula plus what the symbols mean, is a full 2 marks.

76 questions: every 2-mark part asked across the 6 papers, plus every additional 2-marker the MAC21 portion can support. Counts are appearances across those papers; LIKELY means not yet asked but squarely on the syllabus. The two computational entries (#48, #55) were computed and verified. Answers are exam-length by design.